Lesson 08 Demo 07

Validating, Applying, and Destroying the Terraform file

Objective: To validate, apply, and destroy the Terraform configuration file for efficient and reliable management of your infrastructure lifecycle

Tools required: VS Code and Linux terminal

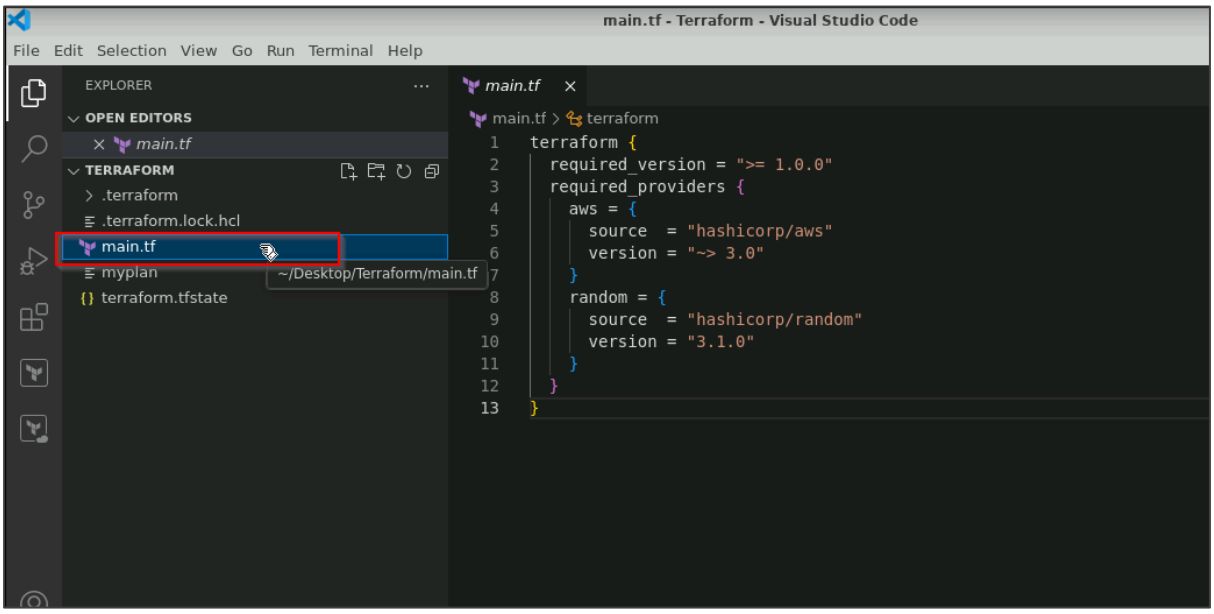
Prerequisites: None

Steps to be followed:

1. Modify the main.tf file with the provider configuration code
2. Validate the Terraform configuration file
3. Apply the validated Terraform configuration file
4. Destroy the applied Terraform configurations

Step 1: Modify the main.tf file with the provider configuration code

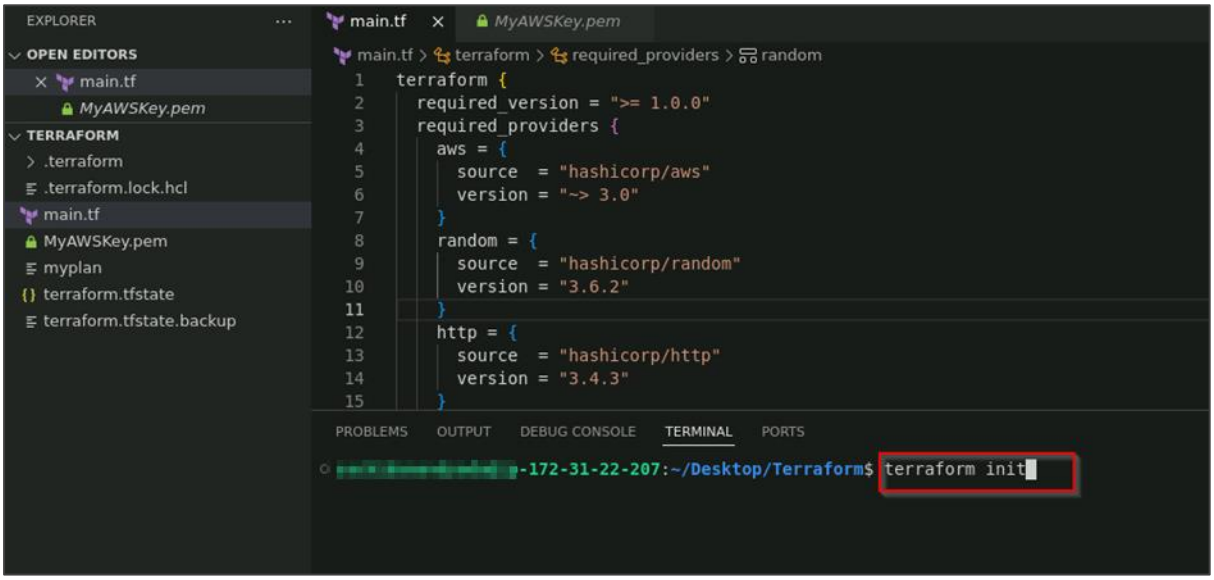
1.1 Open the **main.tf** file

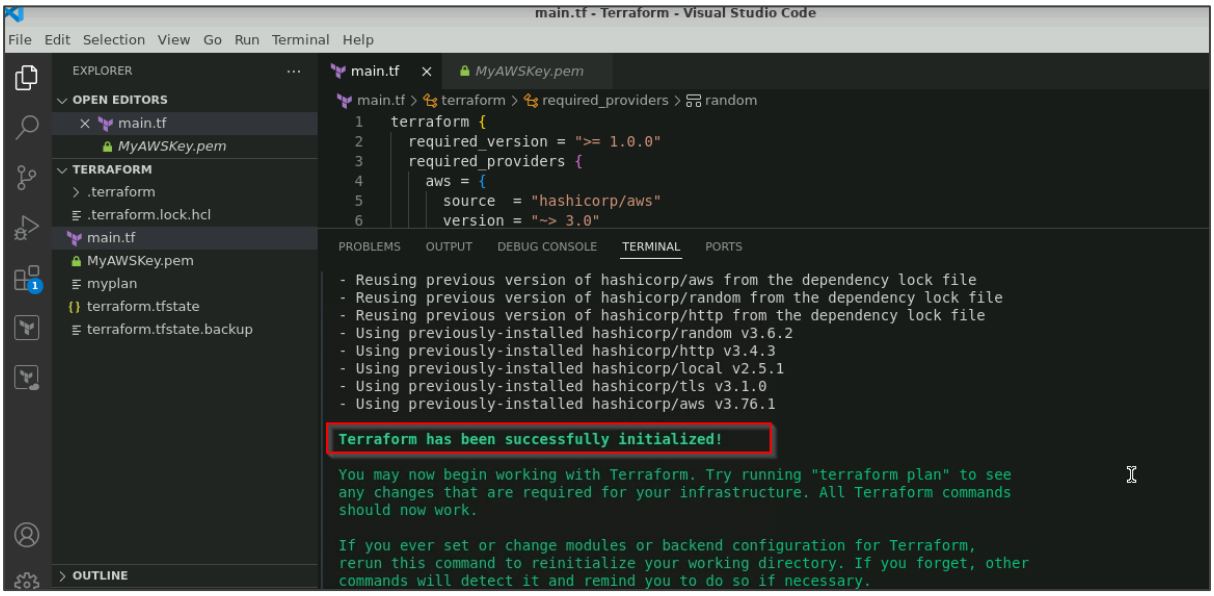


1.2 Modify the **main.tf** file with the code provided below and save the file:

```
terraform {
  required_version = ">= 1.0.0"
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 3.0"
    }
    random = {
      source  = "hashicorp/random"
      version = "3.6.2"
    }
    http = {
      source  = "hashicorp/http"
      version = "3.4.3"
    }
    local = {
      source  = "hashicorp/local"
      version = "2.5.1"
    }
    tls = {
      source  = "hashicorp/tls"
      version = "3.1.0"
    }
  }
}
provider "aws" {
  # Replace with your actual AWS credentials
  access_key = "AKIARJTG7GGYJOVDRU3B"
  secret_key = "kq2lAmP5ajaui+VEhdHMcic4fXmUMcpQM3avt1wD"
  region     = "us-east-1" # Replace with your desired region
}
resource "tls_private_key" "generated" {
  algorithm = "RSA"
}
resource "local_file" "private_key_pem" {
  content  = tls_private_key.generated.private_key_pem
  filename = "MyAWSKey.pem"
}
resource "random_string" "random" {
  length      = 9
  special     = true
  min_numeric = 6
  min_special = 2
  min_upper   = 3
}
```

1.3 Run the command given below to initialize the configurations:
terraform init

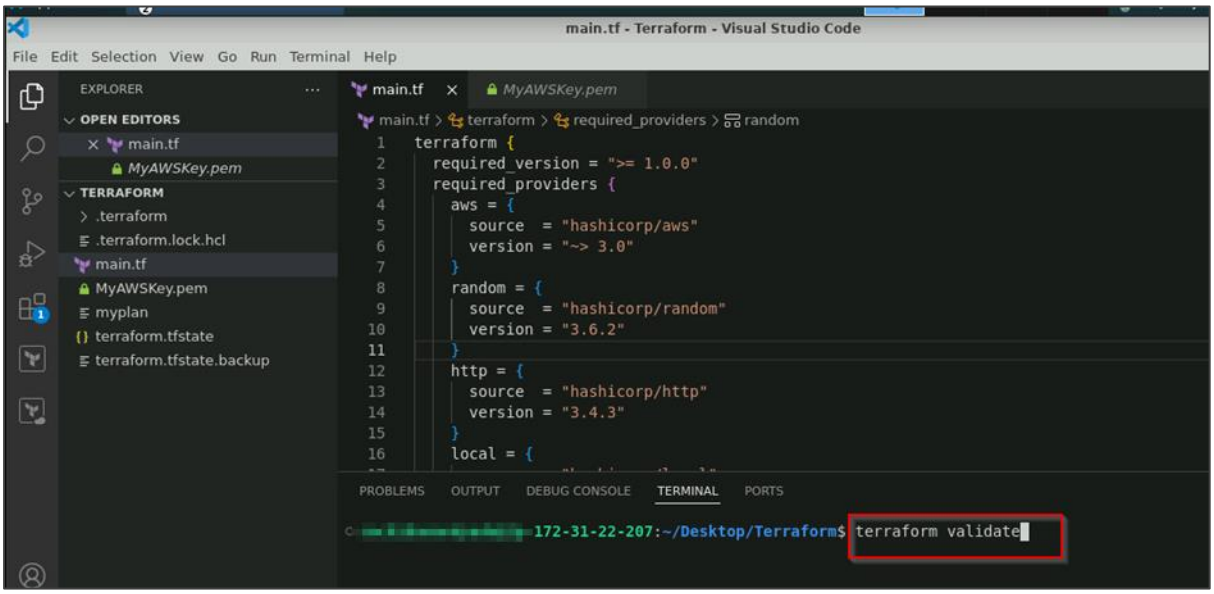




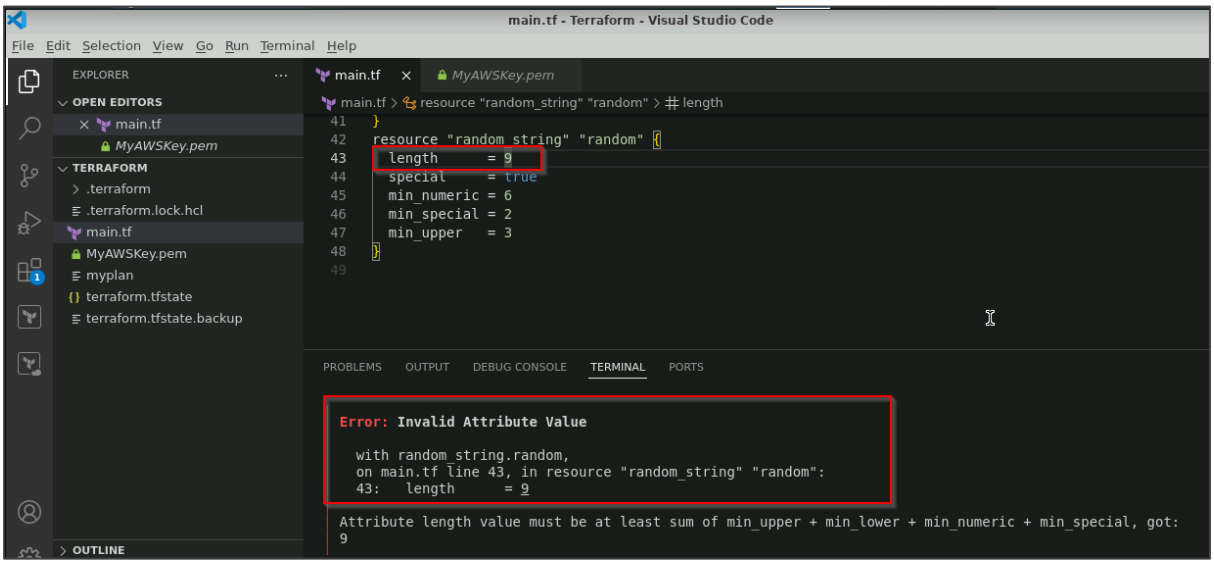
The Terraform file is successfully initialized.

Step 2: Validate the Terraform configuration file

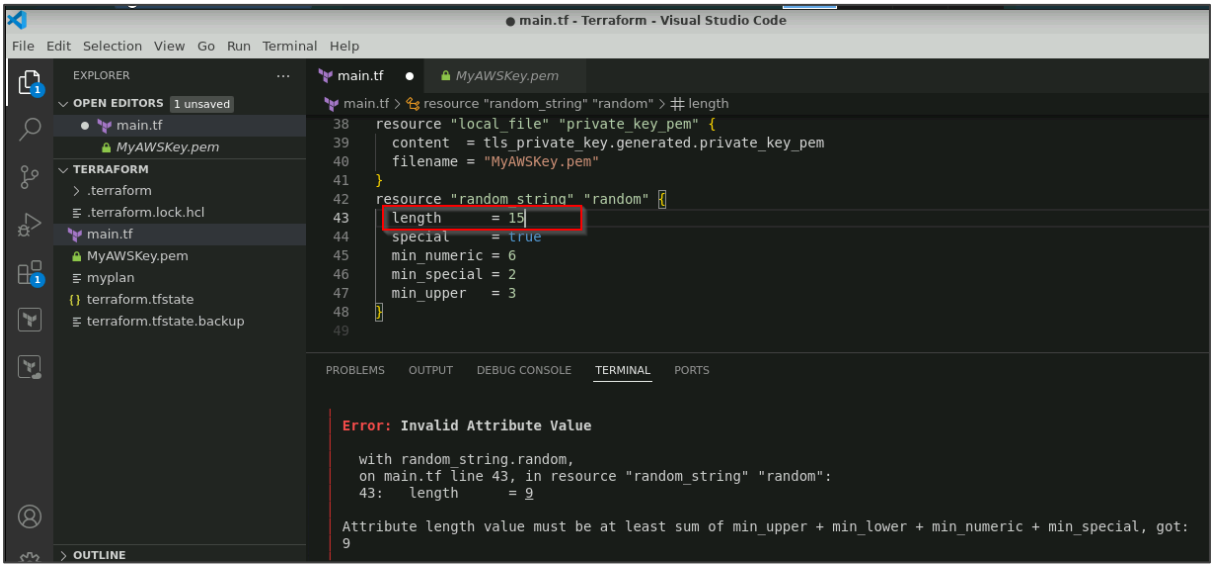
2.1 Open the terminal and execute the command given below to validate the Terraform configuration file:
terraform validate



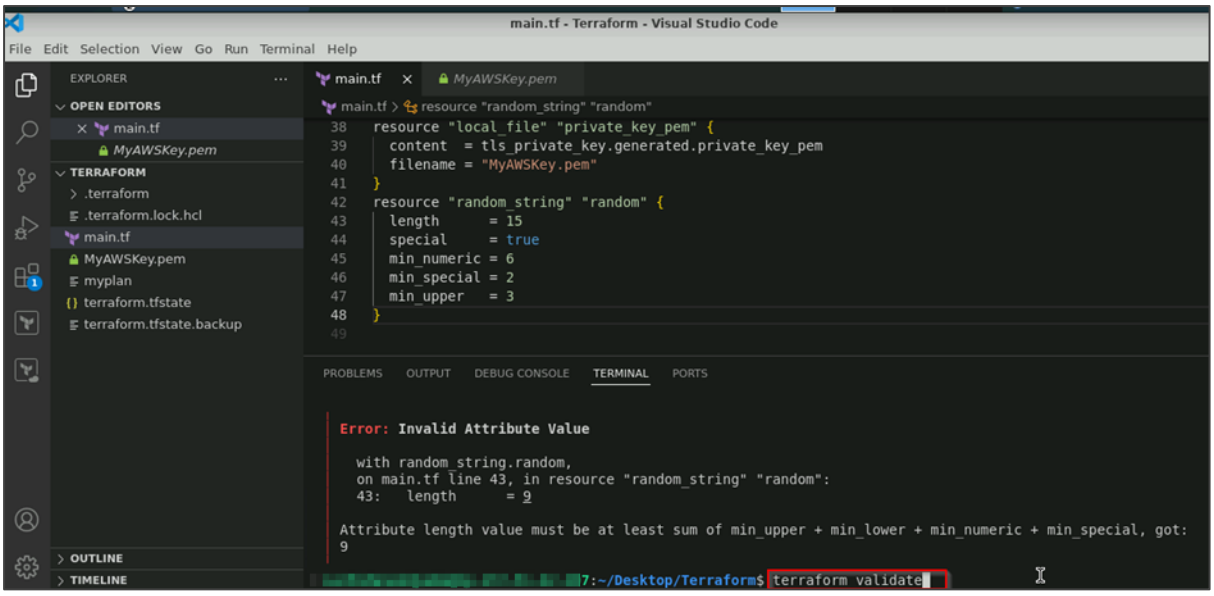
The validate command shows the error in the configuration file and suggests changes.



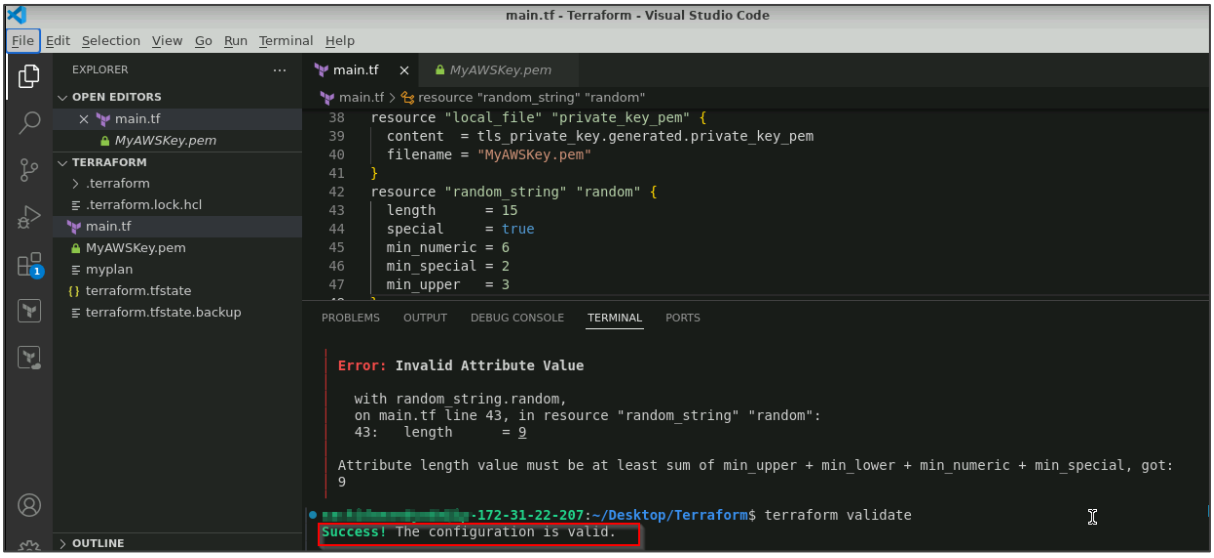
2.2 Make the changes as per the suggestions, replace the value of **length** with **15,** and save the file



2.3 Run the validate command once again to check the validation



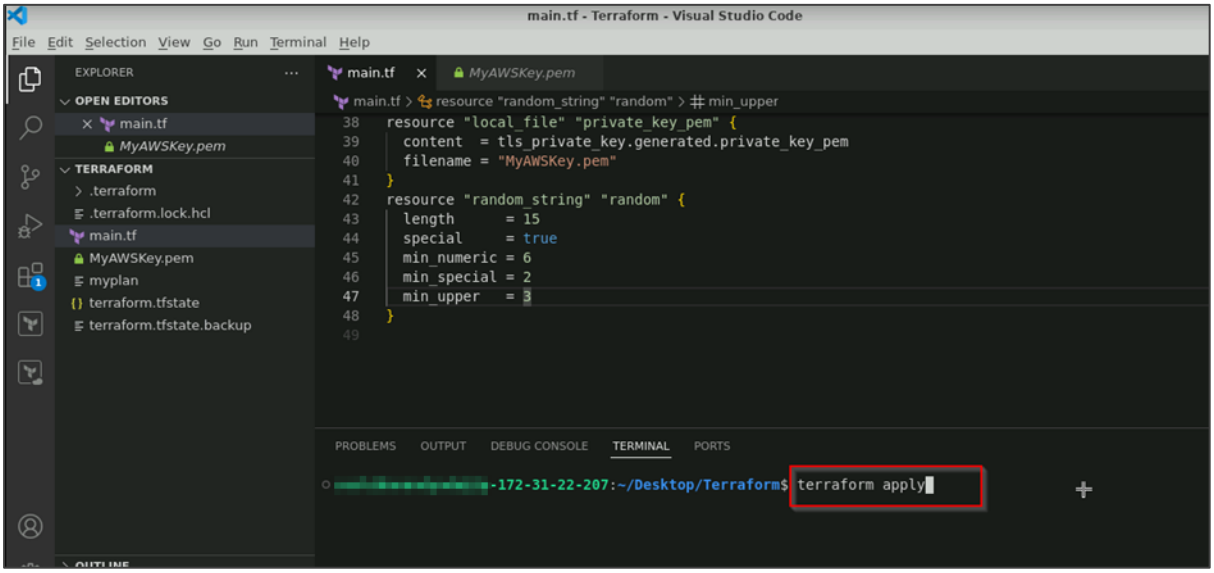
The configuration is valid as the output shows:



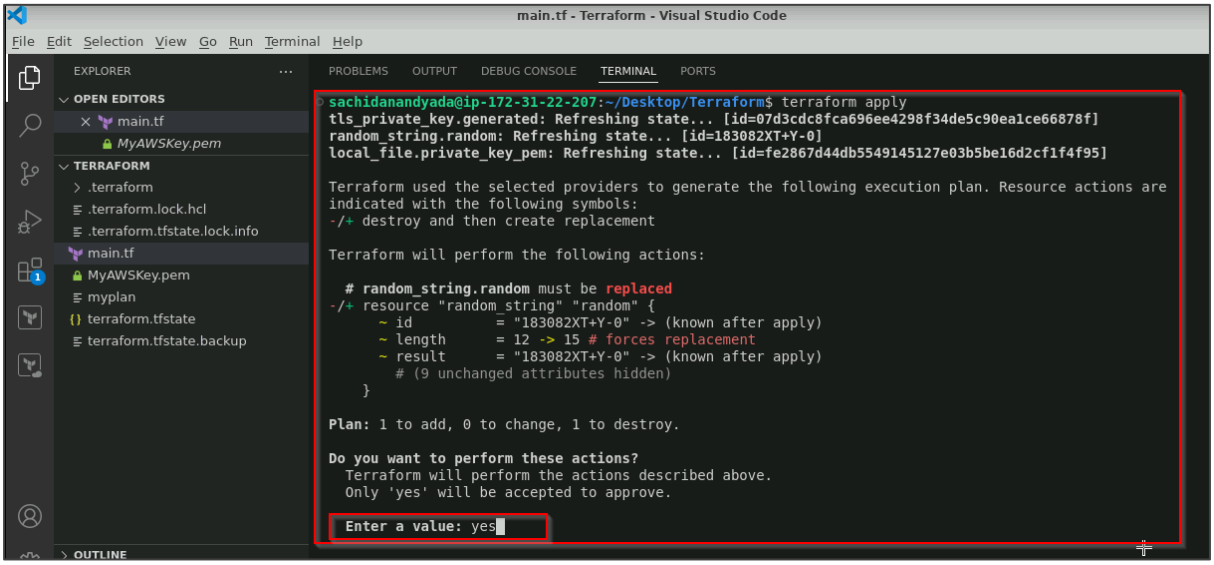
Step 3: Apply the validated Terraform configuration file

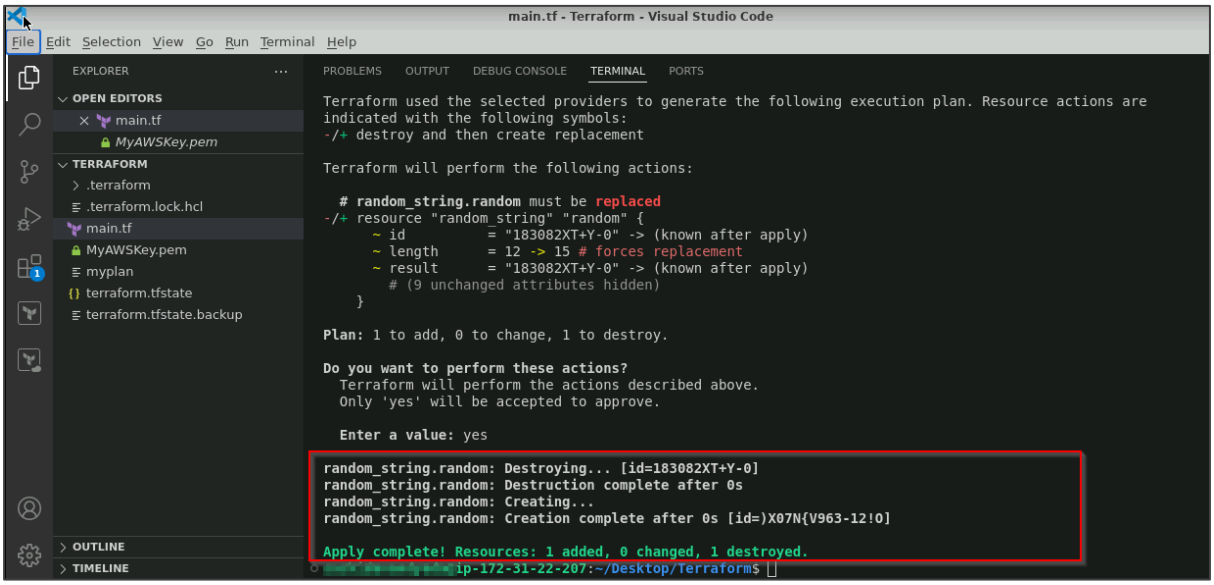
3.1 Open the terminal and enter the command given below to apply the configuration file:

terraform apply



3.2 Confirm with **yes** to apply the configuration

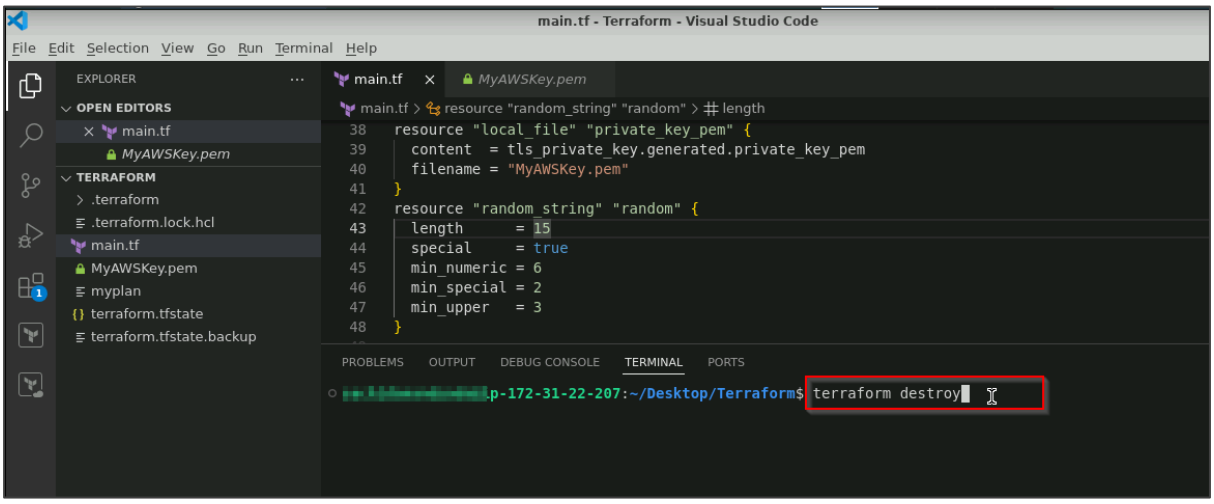




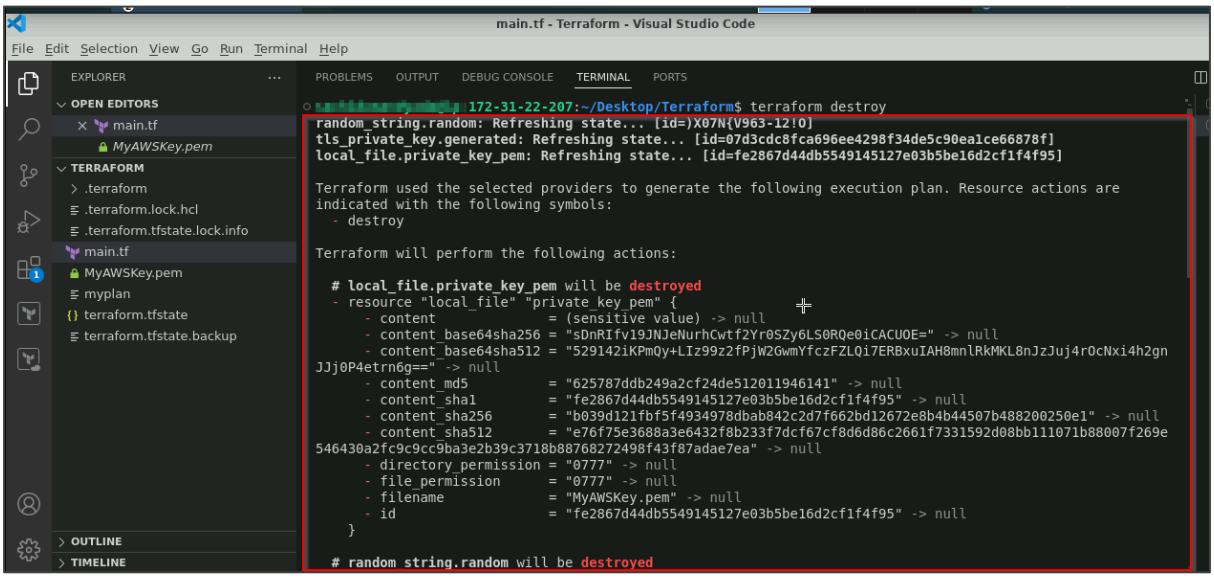
The apply command is completed successfully.

Step 4: Destroy the applied Terraform configurations

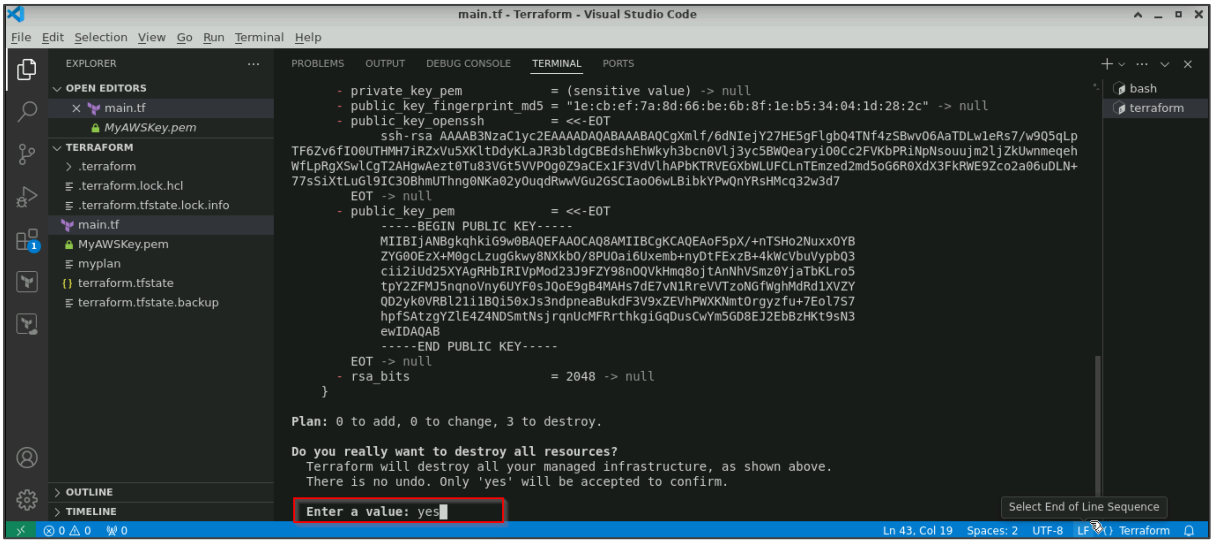
4.1 Open the terminal and execute the given command to destroy the configuration file:
terraform destroy

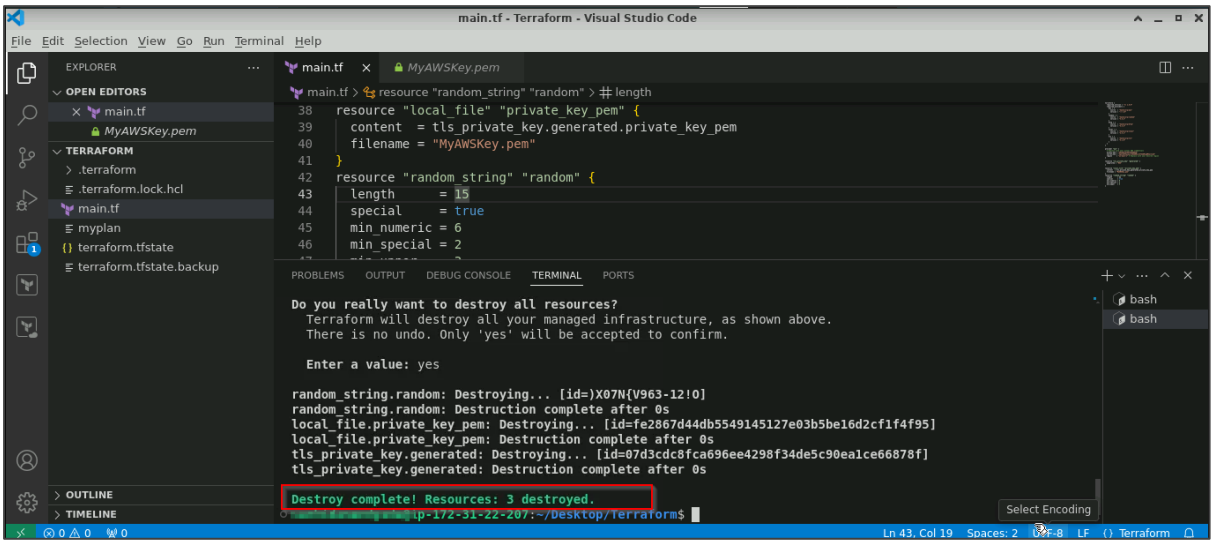


The destroy command will initiate the destruction of the configuration files.



4.2 Confirm with **yes** to destroy the configuration files





The Terraform configuration files are successfully destroyed.

By following these steps, you have successfully validated, applied, and destroyed the Terraform configuration file for efficient and reliable management of your infrastructure lifecycle.